

IIZUKA Tetsuya

associate professor

基本信息

姓名: IIZUKA Tetsuya

所属院校: 东京大学

地区: Japan

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

模拟与数模混合电路

微电子器件与制造

混合信号集成技术

数字集成电路设计

射频集成电路

模拟与射频IC设计

超大规模集成电路设计

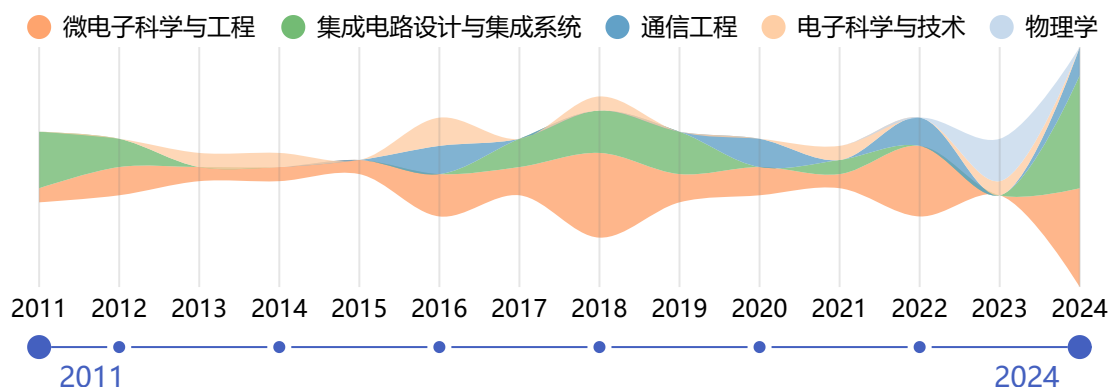
纳米电子与先进工艺

个人简介

Associate Professor IIZUKA Tetsuya specializes in Systems & Electronics at the Hongo Campus, focusing on Computer Systems, Electron Devices, and Communication/Network Engineering. His research emphasizes the design and optimization of integrated circuits and systems, particularly in millimeter-wave integrated circuits for 5G/6G applications, high-precision magnetic field measurement techniques, high-performance analog/digital mixed-signal circuits, time-domain signal processing, and AI and brain-inspired computing. His work aims to enhance system performance and energy efficiency across various technological domains.

工作经历

近年发表的论文



Real-time information processing via volatile resistance change in scalable protonic devices

COMMUNICATIONS MATERIALS

Pati, Satya Prakash; Geng, Yifan; Hamasuna, Satoshi; Fujiwara, Kantaro; Iizuka, Tetsuya; Inoue, Hisashi; Inoue, Isao; Yajima, Takeaki

被引用次数: 0

2024

A 6-GHz 78-fsRMS Double-Sampling PLL With Low-Ripple Bootstrapped DSPD and Retimer-Less MMD Achieving -92-dBc Reference Spur and -258-dB FOM

IEEE MICROWAVE AND WIRELESS TECHNOLOGY LETTERS

Ren, Hongyu; Yang, Zunsong; Huang, Yunbo; Feng, Chaoping; Chen, Tianle; Zhang, Xinming; Meng, Xianghe; Yan, Weiwei; Zhang, Weidong; Iizuka, Tetsuya; Chen, Yong; Mak, Pui-In; Han, Zhengsheng; Li, Bo

被引用次数: 2

2024

Taming Prolonged Ionic Drift-Diffusion Dynamics for Brain-Inspired Computation

ADVANCED MATERIALS

Inoue, Hisashi; Tamura, Hiroto; Kitoh, Ai; Chen, Xiangyu; Byambadorj, Zolboo; Yajima, Takeaki; Hotta, Yasushi; Iizuka, Tetsuya; Tanaka, Gouhei; Inoue, Isao H.

被引用次数: 0

2024

150 GHz Fundamental Oscillator Utilizing Transmission-Line-Based Inter-Stage Matching in 130 nm SiGe BiCMOS Technology

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kano, Sota; Iizuka, Tetsuya

被引用次数: 0

2024

Investigation and Improvement on Self-Dithered MASH $\Delta\Sigma$ Modulator for Fractional-N Frequency Synthesis

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Zhu, Yuyang; Yang, Zunsong; Osada, Masaru; Zhang, Haoming; Iizuka, Tetsuya

被引用次数: 0

2024

A Harmonic-Mixer-Based Fractional-N PLL Employing Voltage-Domain Feed-Forward Noise Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Haoming; Osada, Masaru; Zhu, Yuyang; Iizuka, Tetsuya

被引用次数: 0

2024

A Single Ring-Oscillator-Based Test Structure for Timing Characterization of Dynamic Circuit

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Haoming; Li, Shuowei; Iizuka, Tetsuya

被引用次数: 1

2024

Systematic Equation-Based Design of a 10-Bit, 500-MS/s Single-Channel SAR A/D Converter With 2-GHz Resolution Bandwidth

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Iizuka, Tetsuya; Takenaka, Ritaro; Xu, Hao; Abidi, Asad A.

被引用次数: 0

2024

A Type-II Reference-Sampling PLL with Non-Uniform Octuple-Sampling Phase Detector Achieving 55-fs JitterRMS, -91.9-dBc Reference Spur and -259-dB Jitter-Power FOM

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Ren, Hongyu; Huang, Yunbo; Yang, Zunsong; Chen, Tianle; Meng, Xianghe; Yan, Weiwei; Zhang, Weidong; Liu, Zhongmao; Iizuka, Tetsuya; Mak, Pui-In; Chen, Yong; Liu, Bo

被引用次数: 0

2024

Su-Schrieffer-Heeger Topological Electrical Circuit Using In-Plane Mutual Inductance

2024 SYMPOSIUM ON DESIGN, TEST, INTEGRATION AND PACKAGING OF MEMS/MOEMS, DTIP 2024

Takahashi, Ryohei; Misumi, Kei; Tsuji, Keigo; Eiler, Anne-Claire; Yasunaga, Shun; Higo, Akio; Nakane, Ryosho; Iizuka, Tetsuya; Ezawa, Motohiko; Mita, Yoshio

被引用次数: 0

2024

Guest Editorial: Introduction to the Special Section on the 2023 Asian Solid-State Circuits Conference (A-SSCC)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Takeuchi, Ken; Iizuka, Tetsuya; Nishimura, Kazuko; Yoo, Jerald

被引用次数: 0

2024

A Fractional-N Ring PLL Using Harmonic-Mixer-Based Dual Feedback and Split-Feedback Frequency Division With Phase-Domain Filtering

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Osada, Masaru; Xu, Zule; Yang, Zunsong; Iizuka, Tetsuya

被引用次数: 1

2024

A Unified OTP and PUF Exploiting Post-Program Current on Standard CMOS Technology

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Serrano, Ronaldo; Duran, Ckristian; Sarmiento, Marco; Nguyen, Khai-Duy; Iizuka, Tetsuya; Hoang, Trong-Thuc; Pham, Cong-Kha

被引用次数: 1

2024

An All-Standard-Cell-Based Synthesizable SAR ADC with Inverter-Cell-Based Capacitive DAC

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Li, Shuowei; Yang, Zunsong; Iizuka, Tetsuya

被引用次数: 0

2024

Universal quantum computer based on carbon nanotube rotators

JAPANESE JOURNAL OF APPLIED PHYSICS

Ezawa, Motohiko; Yasunaga, Shun; Iizuka, Tetsuya; Higo, Akio; Mita, Yoshio

被引用次数: 0

2023

Universal quantum computation based on nanoelectromechanical systems

PHYSICAL REVIEW RESEARCH

Ezawa, Motohiko; Yasunaga, Shun; Higo, Akio; Iizuka, Tetsuya; Mita, Yoshio

被引用次数: 0

2023

TopoMEMS varicaps - MEMS comb-drive variable capacitors with tailored stroke-to-capacitance dependence

2023 SYMPOSIUM ON DESIGN, TEST, INTEGRATION & PACKAGING OF MEMS/MOEMS, DTIP

Mita, Yoshio; Yasunaga, Shun; Tsuji, Keigo; Higo, Akio; Iizuka, Tetsuya; Ezawa, Motohiko

被引用次数: 0

2023

Experimental demonstration of position-controllable topological interface states in high-frequency Kitaev topological integrated circuits

COMMUNICATIONS PHYSICS

Iizuka, Tetsuya; Yuan, Haochen; Mita, Yoshio; Higo, Akio; Yasunaga, Shun; Ezawa, Motohiko

被引用次数: 4

2023

A Charge-Redistribution Multi-Bit Stochastic-Resonance ADC Enhancing SNDR for Weak Input Signal

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Shibata, Ryoya; Xu, Zule; Hotta, Yasushi; Tabata, Hitoshi; Iizuka, Tetsuya

被引用次数: 1

2022

140 GHz Energy-Efficient OOK Receiver using Self-Mixer-Based Power Detector in 65nm CMOS

2022 INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY (ICICDT)

Nguyen Ngoc Mai-Khanh; Yamazaki, Daisuke; Iizuka, Tetsuya

被引用次数: 0

2022

Analysis of strong-arm comparator with auxiliary pair for offset calibration

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Li, Shuowei; Xu, Zule; Iizuka, Tetsuya

被引用次数: 1

2022

An ultra-compact leaky integrate-and-fire neuron with long and tunable time constant utilizing pseudo resistors for spiking neural networks

JAPANESE JOURNAL OF APPLIED PHYSICS

Chen, Xiangyu; Yajima, Takeaki; Inoue, Isao H.; Iizuka, Tetsuya

被引用次数: 5

2022

4-Cycle-Start-Up Reference-Clock-Less Digital CDR Utilizing TDC-Based Initial Frequency Error Detection with Frequency Tracking Loop

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Chin, Meikan; Nakura, Toru; Asada, Kunihiro

被引用次数: 0

2022

A fractional-N MASH2-k FDC phase-locked loop architecture enabling higher-order quantisation noise shaping

ELECTRONICS LETTERS

Iwashita, Ryoga; Xu, Zule; Osada, Masaru; Iizuka, Tetsuya

被引用次数: 0

2022

Shock-wave Transceiver Integration for Mm-wave Active Sensing Applications

2021 INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY (ICICDT)

Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2021

A Tutorial on Systematic Design of CMOS A/D Converters: Illustrated by a 10 b, 500 MS/s SAR ADC with 2 GHz RBW

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Iizuka, Tetsuya; Xu, Hao; Abidi, Asad A.

被引用次数: 4

2021

A 40 nm 16 Gb/s differential transmitter with far-end crosstalk cancellation using injection timing control for high-density flexible flat cables

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Takahashi, Daigo; Fujita, Yusuke; Miura, Satoshi; Iizuka, Tetsuya

被引用次数: 0

2020

11 Gb/s 140 GHz OOK modulator with 24.6 dB isolation utilising cascaded switch and amplifier-based stages in 65 nm bulk CMOS

IET CIRCUITS DEVICES & SYSTEMS

Yamazaki, Daisuke; Otsuki, Yoshitaka; Hara, Takafumi; Khanh, Nguyen Ngoc Mai; Iizuka, Tetsuya

被引用次数: 5

2020

A Calibration Technique for Simultaneous Estimation of Actual Sensing Matrix Coefficients on Modulated Wideband Converters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Byambadorj, Zolboo; Asami, Koji; Yamaguchi, Takahiro J.; Higo, Akio; Fujita, Masahiro; Iizuka, Tetsuya

被引用次数: 10

2020

A 16-bit 2.0-ps Resolution Two-Step TDC in 0.18- μ m CMOS Utilizing Pulse-Shrinking Fine Stage With Built-In Coarse Gain Calibration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Enomoto, Ryuichi; Iizuka, Tetsuya; Koga, Takehisa; Nakura, Toru; Asada, Kunihiro

被引用次数: 21

2019

Spatial resolution improvement for point light source detection in scintillator cube using SPAD array with multi pinholes

IEICE ELECTRONICS EXPRESS

Iizuka, Tetsuya; Xu, Kai; Yang, Xiao; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2019

A compact quick-start sub-mW pulse-width-controlled PLL with automated layout synthesis using a place-and-route tool

IEICE ELECTRONICS EXPRESS

Jing, Wang; Iizuka, Tetsuya; Xu, Zule; Nakura, Toru

被引用次数: 0

2019

A 140 GHz area-and-power-efficient VCO using frequency doubler in 65 nm CMOS

IEICE ELECTRONICS EXPRESS

Otsuki, Yoshitaka; Yamazaki, Daisuke; Nguyen Ngoc Mai Khanh; Iizuka, Tetsuya

被引用次数: 1

2019

Triangular Active Charge Injection Method for Resonant Power Supply Noise Reduction

IEICE TRANSACTIONS ON ELECTRONICS

Kano, Masahiro; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2018

Analysis and design of impulse signal generator based on current-mode excitation and transmission line resonator

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Kanjanavirojkul, Parit; Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 0

2018

Comprehensive Analysis of Distortion in the Passive FET Sample-and-Hold Circuit

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Iizuka, Tetsuya; Ito, Takaaki; Abidi, Asad A.

被引用次数: 24

2018

Quick-Start Pulse Width Controlled PLL with Frequency and Phase Presetting

IEICE TRANSACTIONS ON ELECTRONICS

Nakura, Toru; Kagaya, Tsukasa; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2018

A Unified Analysis of the Signal Transfer Characteristics of a Single-Path FET-R-C Circuit

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Abidi, Asad A.

被引用次数: 0

2018

Optimal Design Method of Sub-Ranging ADC Based on Stochastic Comparator

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Hossain, Md Maruf; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 2

2018

Time-domain approach for analog circuits in deep sub-micron LSI

IEICE ELECTRONICS EXPRESS

Asada, Kunihiro; Nakura, Toru; Iizuka, Tetsuya; Ikeda, Makoto

被引用次数: 17

2018

Digitally-Controlled Compensation Current Injection to ATE Power Supply for Emulation of Customer Environment

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Terao, Naoki; Nakura, Toru; Ishida, Masahiro; Ikeno, Rimon; Kusaka, Takashi; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2018

A Gate Delay Mismatch Tolerant Time-Mode Analog Accumulator Using a Delay Line Ring

IEICE TRANSACTIONS ON ELECTRONICS

Yano, Tomohiko; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2017

A PLL Compiler from Specification to GDSII

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2017

Design, Analysis and Implementation of Pulse Generator by CMOS Flipped on Glass for Low Power UWB-IR

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kanjanavirojkul, Parit; Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2017

A 15 x 15 single photon avalanche diode sensor featuring breakdown pixels extraction architecture for efficient data readout

JAPANESE JOURNAL OF APPLIED PHYSICS

Yang, Xiao; Zhu, Hongbo; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 4

2016

FET-R-C Circuits: A Unified Treatment-Part I: Signal Transfer Characteristics of a Single-Path

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Iizuka, Tetsuya; Abidi, Asad A.

被引用次数: 36

2016

FET-R-C Circuits: A Unified Treatment-Part II: Extension to Multi-Paths, Noise Figure, and Driving-Point Impedance

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Iizuka, Tetsuya; Abidi, Asad A.

被引用次数: 38

2016

An Asynchronous Summation Circuit for Noise Filtering in Single Photon Avalanche Diode Sensors

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Yang, Xiao; Zhu, Hongbo; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2016

Spatial Resolution Enhancement for Integrated Magnetic Probe by Two-Step Removal of Si-Substrate Beneath the Coil

IEEE TRANSACTIONS ON MAGNETICS

Mai-Khanh, Nguyen Ngoc; Iizuka, Tetsuya; Nakajima, Shigeru; Asada, Kunihiro

被引用次数: 6

2015

A Structured Routing Architecture for Practical Application of Character Projection Method in Electron-Beam Direct Writing

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Ikeno, Rimon; Maruyama, Takashi; Komatsu, Satoshi; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2014

High-Throughput Electron Beam Direct Writing of VIA Layers by Character Projection with One-Dimensional VIA Characters

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Ikeno, Rimon; Maruyama, Takashi; Komatsu, Satoshi; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 1

2013

A 580 fs-Resolution Time-to-Digital Converter Utilizing Differential Pulse-Shrinking Buffer Ring in 0.18 μm CMOS Technology

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Miura, Satoshi; Yamamoto, Ryota; Chiba, Yutaka; Kubo, Shunichi; Asada, Kunihiro

被引用次数: 6

2012

All-Digital PMOS and NMOS Process Variability Monitor Utilizing Shared Buffer Ring and Ring Oscillator

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 8

2012

Frequency Resolution Enhancement for Digitally-Controlled Oscillator Based on a Single-Period Switching Scheme

IEICE TRANSACTIONS ON ELECTRONICS

Kodama, Kazutoshi; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2012

Cascaded Time Difference Amplifier with Differential Logic Delay Cell

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Nakura, Toru; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 1

2011

All-Digital On-Chip Monitor for PMOS and NMOS Process Variability Utilizing Buffer Ring with Pulse Counter

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Jeong, Jaehyun; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2011

1.0 Ps Resolution Time-to-Digital Converter Based-On Cascaded Time-Difference-Amplifier Utilizing Differential Logic Delay Cells

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Iizuka, Tetsuya; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 2

2011

All-digital ramp waveform generator for two-step single-slope ADC

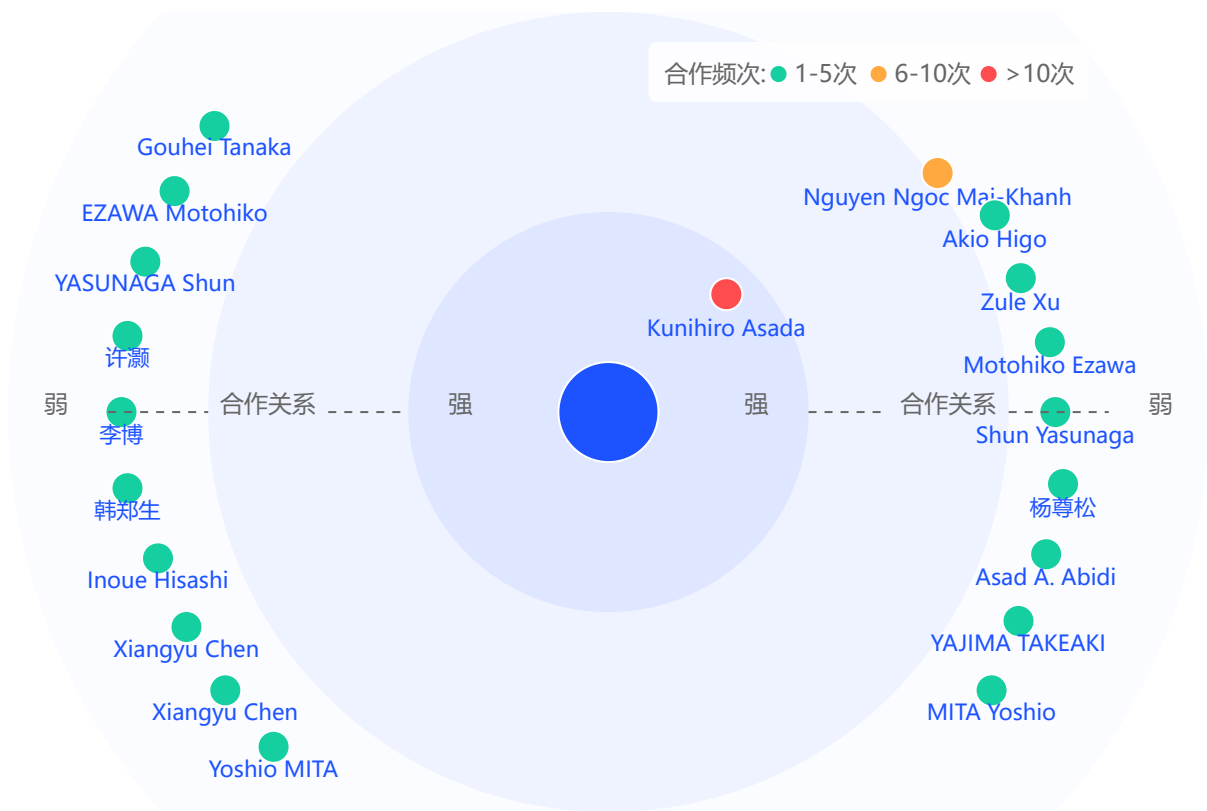
IEICE ELECTRONICS EXPRESS

Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2011

学术合作



Kunihiro Asada

东京大学

合作次数: 25

Nguyen Ngoc Mai-Khanh

东京大学

合作次数: 7

Akio Higo

东京大学

合作次数: 5

Zule Xu

东京大学

合作次数: 5

Motohiko Ezawa

东京大学

合作次数: 4

还有 17 个合作者